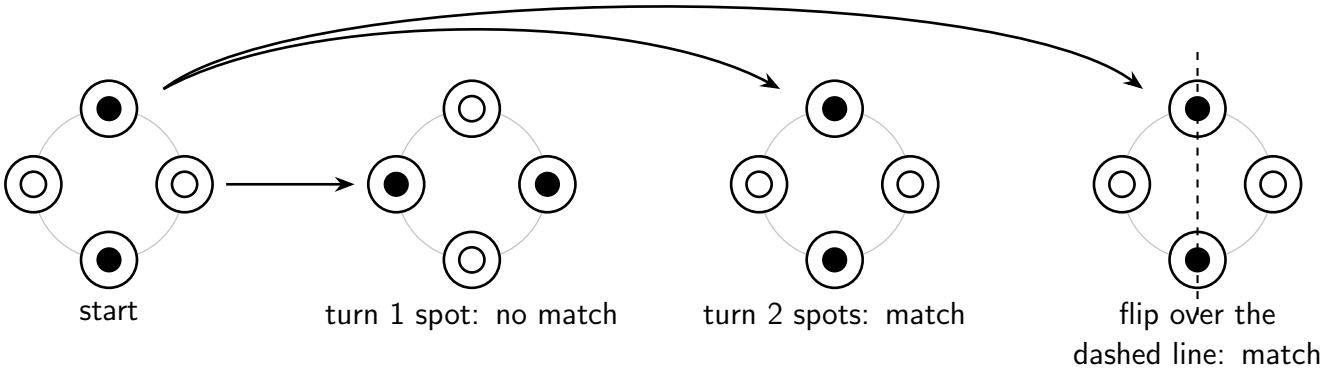
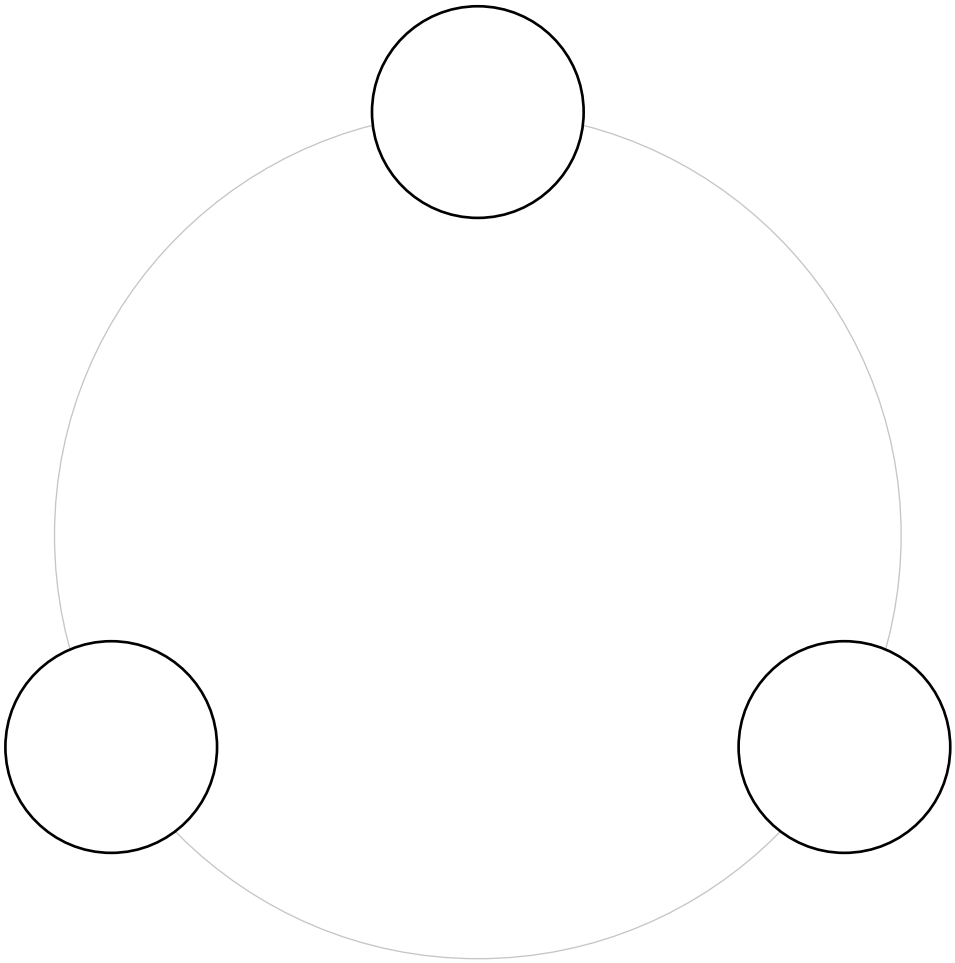
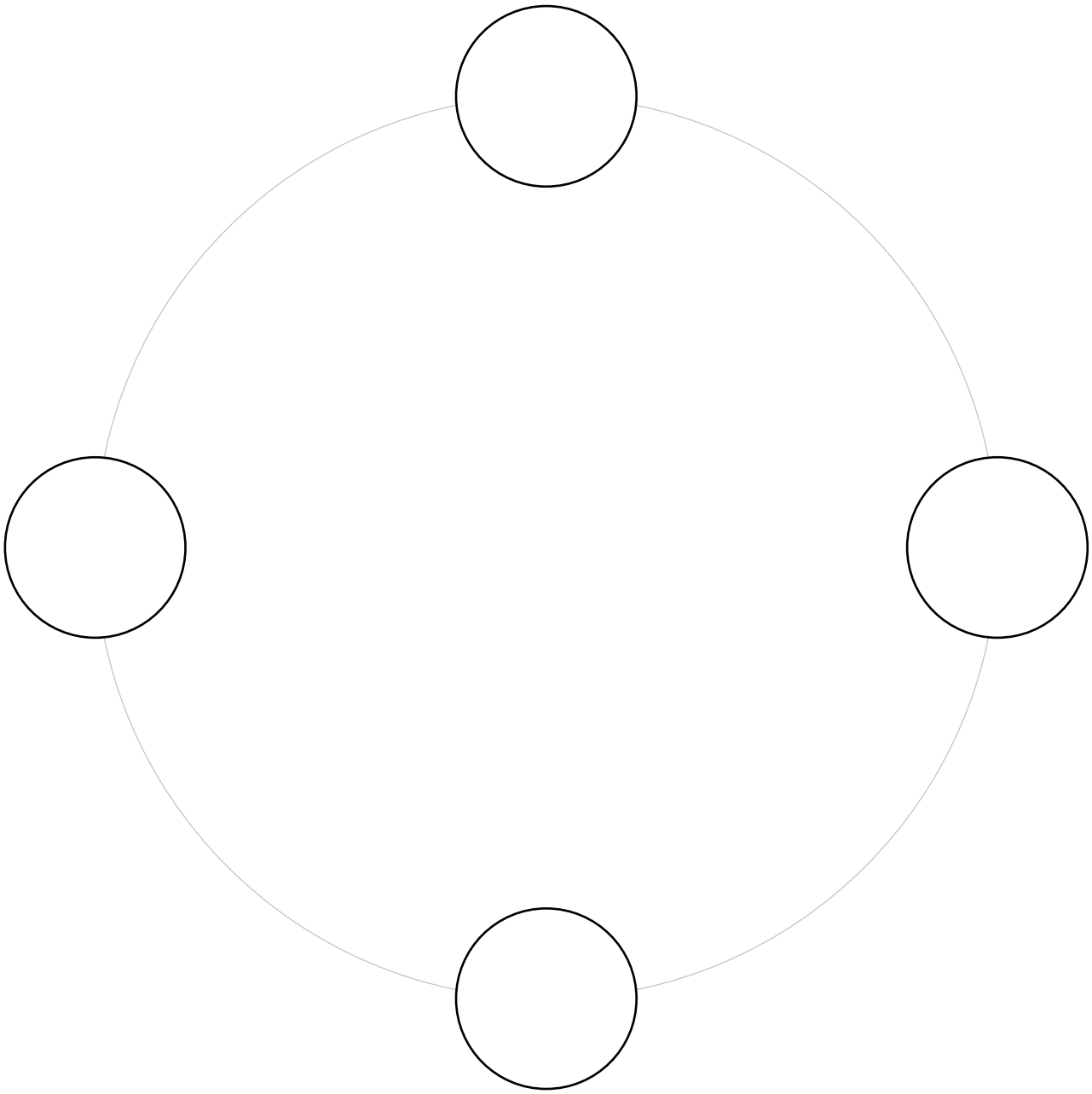
Fill every spot with a counter. Copy the pattern on tracing paper with a dot of each counter's color. One partner builds. The other keeps the counters still and turns or flips (turns over) the copy to a new place. If every dot sits on its own color, that is a match. Putting the copy back where it started, or turning it all the way around, does not count.



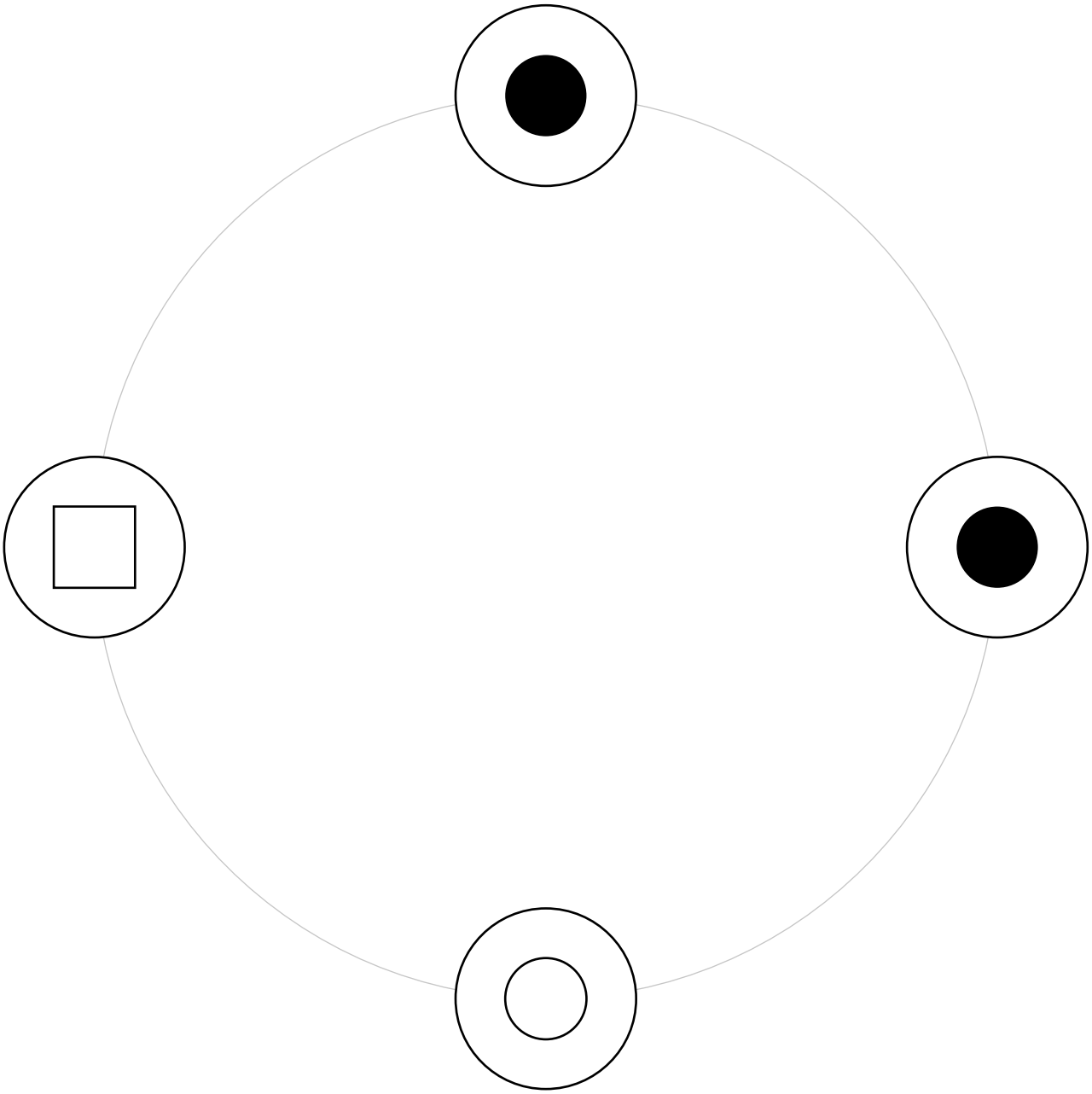
Problem 1: Make different patterns on this ring with two kinds of counter. Can you make one your partner cannot match?



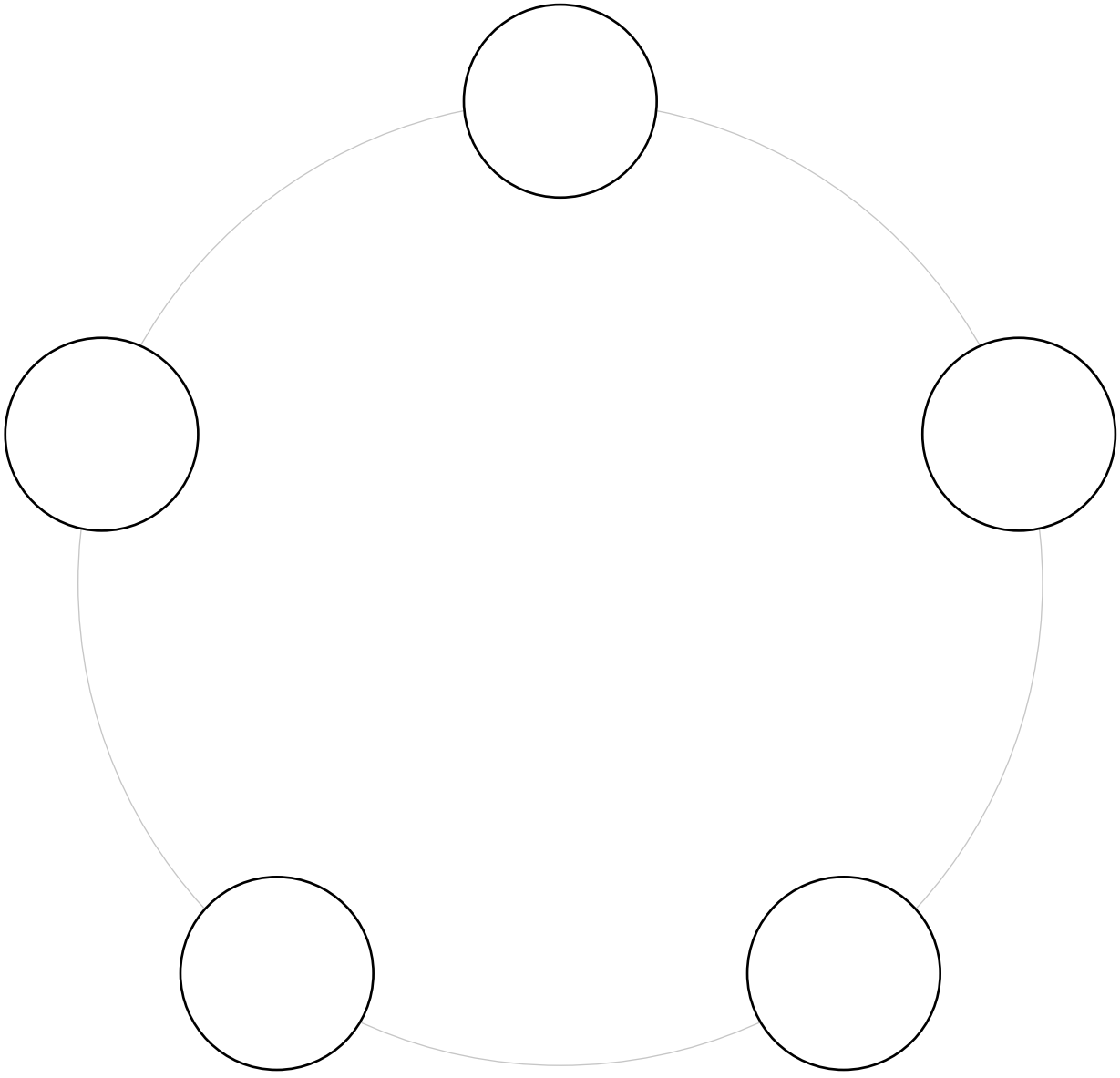
Problem 2: Make a pattern on this ring that no turn or flip can match. How few kinds of counter can you use?



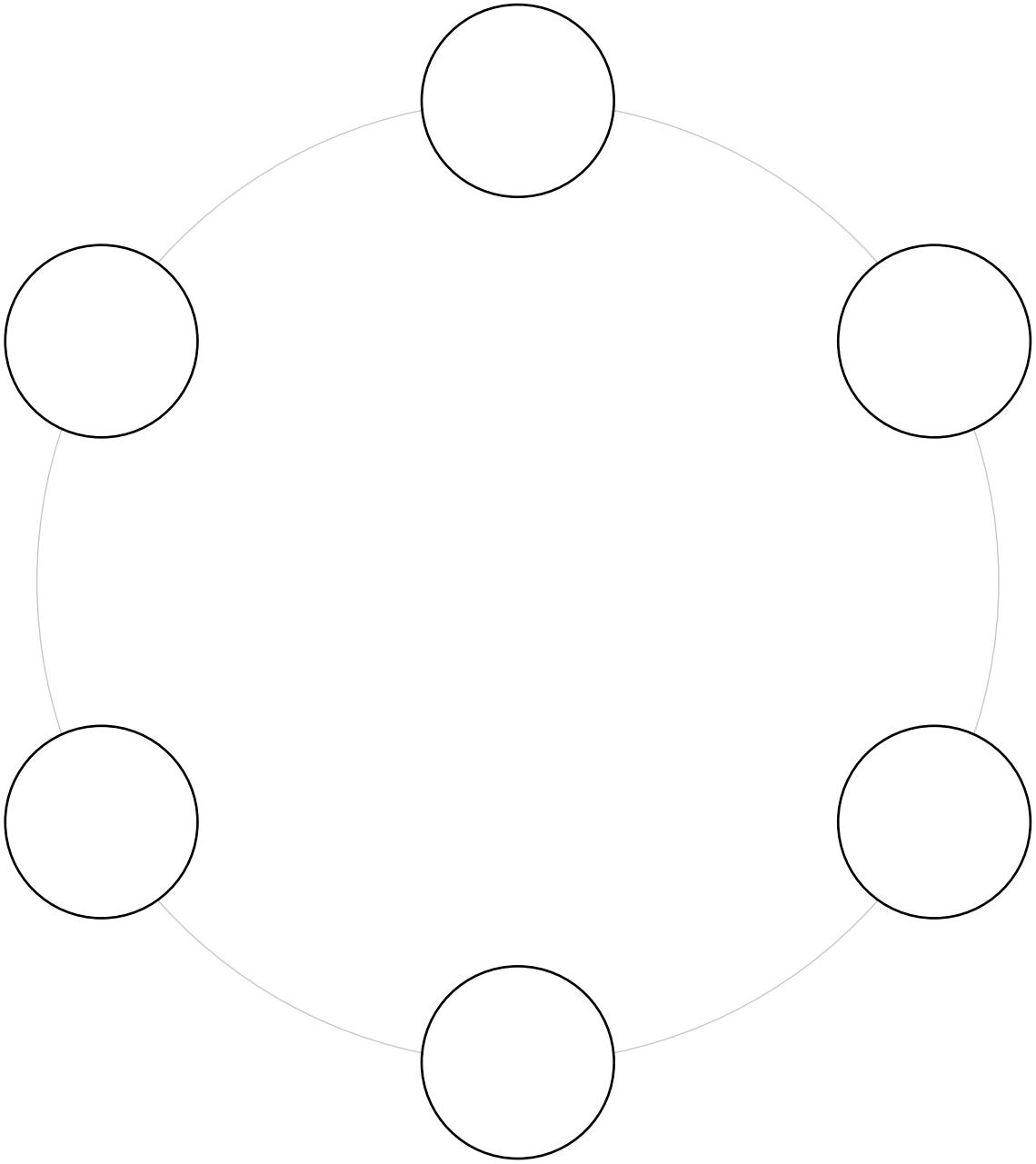
Problem 3: Build this pattern, then change just one counter. Find changes that let a turn or flip match and changes that do not.



Problem 4: Can two kinds of counter make a pattern on this ring that no turn or flip can match? Try three kinds too.



Problem 5: Make a pattern with two kinds of counter that no turn or flip can match.



Problem 6: Make a pattern on this ring with two kinds of counter that no turn or flip can match.

